

Gamification for Learning: Development and Application of Learning Software for Enhancing Student Engagement and Motivation

Shang Wang*
School of Automotive Engineering
Beijing Polytechnic
Beijing, China
wangshang@bpi.edu.cn

Xiangyun Kong
Institute of Engineering Technology
University of Science and Technology Beijing
Beijing, China
supposekong@outlook.com

Nan Wang
School of Automotive Engineering
Beijing Polytechnic
Beijing, China
410913081@qq.com

Abstract—The development of learning software with gamification characteristics has emerged as a solution to enhance students' interest and engagement in learning in vocational colleges. However, designing effective learning software that combines game elements and educational content to overcome students' aversion to formula calculations and improve their learning outcomes poses a significant challenge. To address this challenge, this study focuses on the course of applied mathematics and investigates the development and application of a learning software with game characteristics. The software, developed using the C# language, incorporates a “alarm light” game victory condition and a “mathematical derivation” module to teach system reliability. The effectiveness of the gamified learning software was evaluated through data analysis from a study involving 71 students. The average scores for various questions were as follows: Q1 (2.71), Q2 (2.68), Q3 (2.38), Q4 (2.41), and Q5 (2.54). These results demonstrate the positive impact of the software on enhancing students' learning involvement, motivation, attractiveness of learning content, enjoyment, and their willingness to continue using the software. The development and application of the gamified learning software have proven successful in improving the quality of teaching and students' engagement and motivation in vocational colleges. This highlights the potential of integrating game characteristics into educational software to enhance learning experiences and outcomes.

Keywords—gamification for learning, C# language, learning software, game, engagement, motivation

I. INTRODUCTION

With the rapid advancement of computer and software technologies, the field of education has undergone a transformative shift [1, 2]. Traditional teaching methods, especially in vocational colleges, often fail to captivate students' interest and enthusiasm for learning. Thus, it becomes imperative to develop digitized materials that are easily accessible to vocational college students, such as colorful images, animations, short videos, and mini-games. These interactive resources provide visual representations and foster engagement, making complex concepts more approachable [3, 4]. Integrating these digital assets into learning software can significantly enhance the learning experience and promote a deeper understanding of intricate mathematical principles.

Subjects like applied mathematics, typically taught through conventional classroom methods, tend to elicit aversion and a lack of enthusiasm among students due to their heavy reliance on formula calculations. Consequently, the development of gamified learning software becomes crucial in fostering student engagement and motivation [5]. By infusing game characteristics into the learning process, students are more likely to actively participate and immerse themselves in the subject matter, resulting in a more enjoyable and effective learning experience.

In this study, we have devised learning software that incorporates gamification characteristics, taking “system reliability” as a case study, to facilitate students' comprehension and mastery of relevant concepts and mathematical calculation processes. The learning software has been developed using the C# programming language, effectively integrating the features of series systems and parallel systems by designing game rules and a mechanism for evaluating game results. The outcomes of our teaching experiment have demonstrated that the learning software has created an engaging and challenging learning environment for students.

II. LITERATURE REVIEW

In recent years, there has been a growing interest among scholars in exploring the integration of game characteristics into educational technology. This literature review aims to provide a comprehensive overview of the research conducted by scholars in this field. Haendler and Neumann organized the Se-Games workshop, which focused on games for software engineering education and training [6]. Sure and Wang developed a deep reinforcement learning agent for general video game AI framework games [7]. Gordillo et al. compared the usefulness of game-based learning and video-based learning for teaching software engineering in online environments [8]. Rathnayake et al. investigated the adjustment of game difficulty levels based on predictive models to enhance attraction and business value [9]. López-Fernández et al. examined the learning and motivational impact of game-based learning in face-to-face and online formats, specifically in computer science education [10].

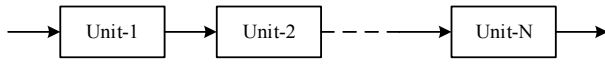


Fig. 1. Reliability diagram of series system.

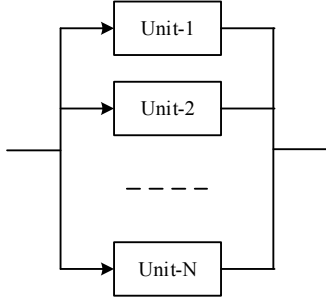


Fig. 2. Reliability diagram of parallel system.

Sharif and Ameen evaluated game engines for serious game development in the context of education [11]. In addition to the integration of game characteristics, developing relevant software for different subjects to improve teaching quality has also become a research hotspot in the education field. Maher et al. put forth an integrated model for learning analytics in adaptive gamified learning, highlighting the importance of visual analytics in enhancing the learning experience [12]. Muzaffar et al. conducted a systematic review of online exam solutions in e-learning, exploring various techniques and tools employed worldwide [13]. Rakic et al. assessed student performance on their e-learning platforms, providing insights into the efficacy of such platforms in improving learning outcomes [14]. Naveed et al. evaluated and ranked critical success factors (CSFs) for cloud-based learning using a combinatorial approach, offering insights for designing optimized learning environments [15]. Wang et al. developed learning software with digital materials based on simulation technology, providing learners with a convenient way to learn abstract mechanical knowledge [16, 17]. Qomariyah and Fajar developed a personalized learning recommender system based on learning styles [18]. Naim et al. emphasized the importance of interoperability in their learning systems and quality development in education [19]. The studies conducted by scholars highlight the potential benefits and effectiveness of game-based learning and adaptive learning software and learning system. These findings contribute to the advancement of educational technology and provide insights for designing innovative and engaging learning environments.

III. METHODS

A. Research Logic

Our research was meticulously conducted in five distinct stages, with a comprehensive focus on the development and investigation of the learning software's profound impact on enhancing learners' motivation within the realm of applied mathematics courses. In the initial phase, we purposefully selected "system reliability" as the prominent case study, devising the fundamental logic and rules that underpin the game. Subsequently, we devised an ingenious random function that effectively encapsulates the notion of "reliability" for individual units, thus enriching the gameplay experience. Additionally, we meticulously designed the average number of "refreshes"

attained by all students in the class, serving as the key criterion for evaluating the overall system reliability within the game. In the third phase, utilizing the C# programming language, we successfully developed the learning software, ensuring its seamless integration with the educational environment. Having accomplished these pivotal milestones, we proceeded to implement the meticulously designed teaching experiment, meticulously monitoring its progress and outcomes. Finally, to gain further insights into the efficacy of our developed learning software, we conducted a comprehensive survey through the distribution of thoughtfully crafted questionnaires, enabling us to glean valuable feedback from the participants. This systematic and rigorous research approach ensures a comprehensive investigation into the profound impact of the developed learning software on learners' motivation within the realm of applied mathematics.

B. Concept of Reliability

The concept of system reliability refers to the ability of a system to fulfill its designated function under specified conditions and within a defined timeframe. It serves as a crucial metric for assessing system performance, particularly for systems that require long-term continuous operation or undertake critical tasks. The reliability of a system depends on the reliability of its constituent components and the manner in which these components are interconnected. Each component carries a probability of failure, which may result in a degradation of system performance or even complete system failure. Therefore, when designing and evaluating systems, careful consideration must be given to the failure probabilities of components and their impact on system reliability. A series system refers to a system where all components are sequentially connected, forming a single path. In a series system, the failure of any component leads to the failure of the entire system. Hence, the reliability of a series system is dependent on its least reliable component. In other words, the reliability of a series system is the product of the reliabilities of its individual components. Fig. 1 demonstrates the reliability of a series system. The overall reliability calculation of a series system is shown in (1).

$$R(t) = \prod_{i=1}^n R_i(t) \quad (1)$$

On the other hand, a parallel system refers to a system where multiple paths exist, with each path composed of different components. In a parallel system, the entire system fails only when all paths experience failures. Therefore, the reliability of a parallel system is typically higher than that of a series system. However, it is important to note that while parallel systems often exhibit higher reliability, they also introduce potential challenges. For instance, if multiple components are simultaneously operational, it may increase energy consumption and system complexity. Additionally, coordination and control issues among components need to be considered. Fig. 2 demonstrates the reliability of a parallel system. The overall reliability calculation of a parallel system is shown in (2).

$$R(t) = 1 - \prod_{i=1}^n [1 - R_i(t)] \quad (2)$$

In conclusion, system reliability is a complex and critical consideration that depends on the combined effects of various

factors. When designing and evaluating systems, it is essential to take into account the reliability of components, the interconnection scheme, and the overall performance requirements of the system.

C. Game Design for Series Systems

To enhance students' understanding of the reliability of series systems, we have developed a game based on a series circuit. This game consists of a circuit with 10 components, each with a reliability value set at 0.91, representing a 91% probability of functioning correctly. The objective of the game is to illuminate a light bulb, which only occurs when all components are operational. This design aims to simulate the real-world behavior of series systems, where the failure of any component leads to the failure of the entire system.

During the software development process, we incorporated a random number generator to simulate the working status of each component. With each program execution, a random number between 0 and 100 is generated for each component. If the generated random number is equal to or below 91, the component is considered to be functional. Conversely, if the random number exceeds 91, the component is deemed to have failed. The light bulb will only illuminate if all 10 components in the circuit are functioning properly, as indicated by the generated random numbers being equal to or below 0.91.

D. Game Design for Parallel Systems

To further aid students in comprehending the reliability of parallel systems, we have developed a game based on a parallel circuit. This game consists of a circuit with 10 components, each with a reliability value set at 0.21, representing a 21% probability of functioning correctly. In this game, the objective is to illuminate a light bulb (same as series systems), which occurs when at least one component in the circuit is operational. This design aims to simulate the behavior of real-world parallel systems, where the failure of one component does not result in the failure of the entire system.

During the software development process, we incorporated a random number generator to simulate the working status of each component. With each program execution, a random number between 0 and 100 is generated for each component. If the generated random number is equal to or below 21, the component is considered to be functional. Conversely, if the random number exceeds 21, the component is deemed to have failed. The light bulb will illuminate if at least one component in the circuit is functioning properly, as indicated by the generated random numbers being equal to or below 0.21.

E. Rules for Playing the Game

In order to enhance the interactivity and appeal of the game, we have implemented a timer and input functionality for tracking the total number of game executions. The reliability of the system is assessed by comparing the frequency of the light bulb illuminating to the total number of experiments conducted. This game offers two distinct modes, which will be described separately.

Firstly, students can simply click the "Run" button to initiate a random execution and observe whether the light bulb illuminates. Regardless of the outcome of the initial attempt,

students have the option to click the "Run" button again to see if the subsequent attempt is successful. This process can be repeated to terminate the execution of tasks. If, after N attempts, students successfully illuminate the light bulb (i.e., win the game), we can express the reliability using (3).

$$R(t) = \frac{1}{N} \quad (3)$$

It should be noted that this reliability metric exhibits a certain degree of randomness. However, as the number of samples increases, corresponding to a larger pool of students (n) participating in the game, the average reliability of the students gradually approaches the theoretical data, as shown in (4).

$$R(t) = \frac{1}{n} \sum_{k=1}^n \frac{1}{N_k} \quad (4)$$

The second mode of the game enables students to directly input a larger number, which determines the number of game executions for calculating the reliability. For instance, a student can input 1000, resulting in the game software executing the operation 1000 times. If the light bulb illuminates a total of 152 times, it indicates a reliability of only 0.152.

Although the results obtained from this game software may deviate from the theoretically derived outcomes, particularly when the number of experiments is limited, this approach provides students with a more intuitive understanding of the variations in reliability control within series-parallel systems. Moreover, students can gain a deeper appreciation that reliability truly manifests itself through a substantial amount of data (number of experiments, total samples).

F. Mathematical Derivation

Through the implementation of the game design discussed earlier, our software effectively aids learners in grasping the concept of system reliability and understanding the influence of series and parallel connections on system stability. While the game serves as a valuable tool for comprehension, it is crucial for students to also acquire a solid understanding of the relevant formulas and mathematical calculation methods. To address this, we meticulously derived and elucidated the underlying mathematical principles within the context of the game design.

In the case of a series system, the failure of any single component results in a complete system failure. Consequently, for the system to be deemed reliable and operational, all 10 components must be functioning flawlessly. By employing (1) and performing the corresponding calculations, we obtained a reliability value of 0.3894 for the series system, as indicated in (5).

$$R(t) = (0.91)^{10} = 0.3894 \quad (5)$$

This value clearly demonstrates that a series system, composed of multiple components, exhibits significantly lower reliability compared to that of an individual component. Similarly, we conducted mathematical calculations for a parallel system.

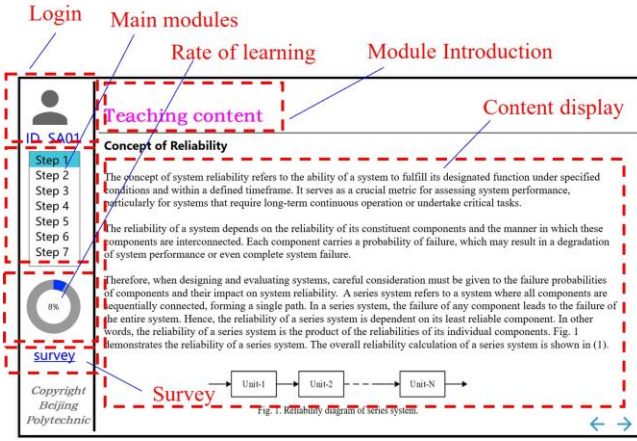


Fig. 3. Learning software interface.

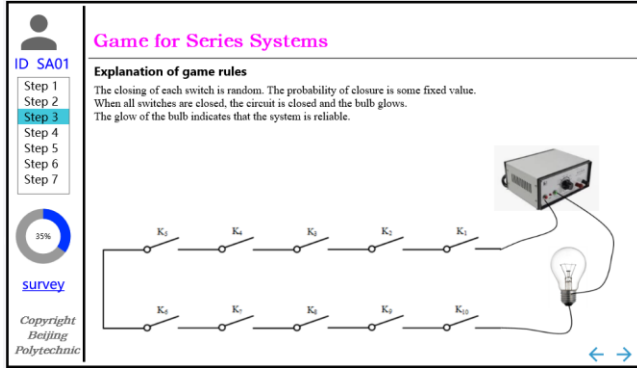


Fig. 4. Interface of game for series systems.

In a parallel circuit, the system functions as long as at least one component is operational, thereby allowing current to flow and illuminating the light bulb in our software. Applying (2) and conducting the necessary calculations, we arrived at a reliability value of 0.9053 for the parallel system, as presented in (6).

$$R(t) = 1 - (1 - 0.21)^{10} = 0.9053 \quad (6)$$

The findings underscore the fact that a parallel system, comprising multiple components, exhibits substantially higher reliability compared to that of an individual component.

The seamless integration of theoretical concepts and practical application fosters a comprehensive learning experience, enhancing student engagement and promoting higher levels of comprehension. It is worth noting that the reliability values obtained through mathematical calculations align closely with the results derived from the gameplay within our software. This congruence reinforces the validity and accuracy of our software's educational approach. The successful combination of game-based learning and mathematical analysis offers a promising avenue for enhancing students' interest, motivation, and learning outcomes in vocational colleges.

G. Learning Software

When developing learning software for mechanics courses, C# becomes an ideal choice due to its powerful capabilities and extensive functionalities. This modern, statically-typed

language ensures efficiency, reduces runtime errors, and simplifies development.

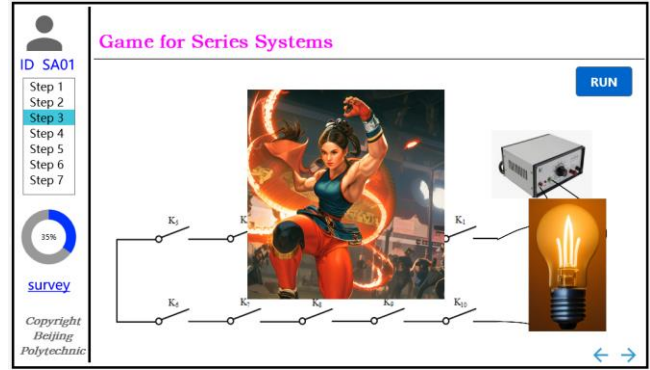


Fig. 5. Interface after winning the game.

Following the previously mentioned game rules, we have developed a gamified learning software. The main interface of the software is shown in Fig. 3. On the left side, there is a navigation bar with functions such as "Registration," "Main Steps," "Learning Progress," and "Assessment," facilitating students' learning process. On the right side, the main content presentation area includes titles and main content presented with images and text. Fig. 4 displays the interface of the "Series Circuit" game. Consistent with the previous description, series circuits are represented using serial circuits. When the light bulb on the right side is lit, it indicates that the system is reliable. Only when the light bulb is lit can the game be considered a victory, and a "Victory" interface, as shown in Fig. 5, will pop up. To enhance the gaming experience for students, we have designed a popular cartoon game image on this interface.

H. Teaching Experiment

By implementing these nine steps in the teaching experiment (as shown in Tab.1), students will attain a comprehensive understanding of system reliability in series systems while acquiring proficiency in operating the accompanying software. Through independent exploration and group discussions, students will deepen their comprehension and application abilities related to system reliability. Moreover, teachers can evaluate the software's effectiveness in facilitating learning, drawing insights from student feedback and experimental results to inform future teaching enhancements.

I. Assessment Method

To evaluate the effectiveness of the gamified learning software, a data analysis methodology was employed. The survey participants consisted of 71 students from vocational colleges. The survey questionnaire comprised five questions (Q1-Q5), which were as follows:

Q1: How effective do you perceive this gamified learning software in enhancing your level of engagement in learning? Q2: After using this learning software, how do you perceive the improvement of your learning motivation? Q3: In your opinion, how does the incorporation of game elements in this learning software impact the appeal of the learning content? Q4: Did you find learning to be more enjoyable and challenging while utilizing this learning software? Q5: Are you willing to continue using this gamified learning software for future learning?

TABLE I. STEPS AND CONTENT OF THE TEACHING EXPERIMENT

The process of teaching experiment.	
Steps	Main content
1	Introduction and Software Overview (5 minutes) Begin by providing an introduction to the concept of series systems and emphasize their significance in practical applications. Briefly introduce the developed software and its objectives, showcasing its interface and functionalities.
2	Explanation of System Reliability Concept (10 minutes) Thoroughly explain the definition of system reliability and its relevance in engineering. Illustrate with examples the impact of component failures on the overall system in series systems.
3	Software Demonstration and Operational Guidance (10 minutes) Demonstrate the step-by-step process of operating the software, including inputting the total number of executions, initiating the simulation, and observing the results. Provide clear instructions on how to properly utilize the software, including data input techniques and result interpretation.
4	Independent Practice by Students (15 minutes) Allocate time for students to independently operate the software and conduct simulation experiments, adhering to the provided guidelines. Teachers should circulate among students to offer guidance and address any queries that arise during the process.
5	Mid-class Break (10 minutes) Designate a break for students to rest while teachers prepare for the subsequent phase of the lesson.
6	Group Discussion and Knowledge Sharing (25 minutes) Organize students into groups to facilitate discussions on the experimental outcomes and encourage the sharing of observed system reliability data. Teachers should actively participate in these discussions, guiding students in analyzing the underlying causes and influencing factors behind the data.
7	Summary and Feedback (10 minutes) Summarize the importance of system reliability in series systems and its associated influencing factors. Solicit feedback from students to assess the efficacy of the software in facilitating learning.
8	Class Review and Question-and-Answer Session (10 minutes) Recapitulate the key content and learning objectives covered in the class. Address any questions or concerns raised by students, ensuring their comprehension of system reliability in series systems.
9	Assignment of Post-Class Exercises (5 minutes) Assign relevant exercises to students, requiring them to apply their acquired knowledge in analyzing different types of series systems. Encourage students to explore additional features and application scenarios of the software beyond the classroom setting.

Each question was rated on a scale of 0 to 3, with 0 representing the lowest rating and 3 representing the highest rating. The participants were instructed to provide their scores to two significant figures. The average scores of the students were calculated, and this data was utilized to evaluate the efficacy of the learning software.

IV. RESULTS

A. Data Analysis

We conducted statistics on the students' ratings, and the results are shown in Tab. 2. Q1 evaluates the effectiveness of the gamified learning software in improving students' level of engagement in learning.

TABLE II. STATISTICS FOR LEARNING SOFTWARE RATINGS.

Data survey results of the students					
Score	Q1	Q2	Q3	Q4	Q5
S	2.71	2.68	2.38	2.41	2.54

The average score of 2.71 indicates that the software has a significant positive effect on student engagement, as the score is above the midpoint of the scale (1.5). This implies that a lot of students perceive the software to be effective in enhancing their learning involvement. Q2 assesses the enhancement of students' learning motivation after using the gamified learning software. With an average score of 2.68, it suggests that the software has a positive impact on increasing students' motivation to learn. The score is above the midpoint, indicating that students have experienced a noticeable improvement in their learning motivation. Q3 investigates the influence of game elements in the learning software on the attractiveness of the learning content. The average score of 2.38 suggests that the inclusion of game elements has a moderately positive impact on the attractiveness of the learning content. Q4 examines whether students found learning to be more enjoyable and challenging while using the gamified learning software. With an average score of 2.41, it indicates that the software has a positive effect on making learning more enjoyable and challenging for students. The score surpasses the midpoint, demonstrating that students perceive an increase in the enjoyment and challenge of learning through the gamified approach. Q5 explores students' willingness to continue using the gamified learning software for future learning. The average score of 2.54 indicates a positive response, as it is above the midpoint. This suggests that students are generally willing to continue using the software for their learning needs, highlighting their satisfaction and engagement with the gamified approach.

Based on the analysis of the data, it is evident that the gamified learning software has a significant positive impact on the quality of teaching and students' level of engagement and motivation. Further exploration and research in this area are warranted to enhance the understanding and implementation of gamified learning approaches in vocational colleges.

B. Shortcomings

The paper could be improved by providing a more comprehensive discussion of the specific features and mechanics of the gamified learning software, which are crucial for understanding its effectiveness and potential limitations. Additionally, the methodology for data collection and analysis should be further elaborated, including details on the sample selection process and the statistical techniques employed.

C. Future Work

Our future research should focus on conducting a more in-depth analysis of the gamified learning software, exploring its specific features and mechanics to gain a deeper understanding of how they contribute to student engagement and motivation. The methodology for data collection and analysis should be refined, providing a more detailed description of the sample selection process, and incorporating additional statistical techniques to enhance the robustness of the findings. Lastly,

extending the research to other vocational college courses and subjects would allow for the assessment of the generalizability of the findings and the exploration of the potential benefits of gamified learning software in different educational contexts.

V. CONCLUSION

The development and application of the gamified learning software in vocational colleges have demonstrated significant positive effects on the quality of teaching and students' level of engagement and motivation. The data analysis reveals that the software effectively enhances students' learning involvement ($S=2.71$), motivation ($S=2.68$), attractiveness of learning content ($S=2.38$), enjoyment, and challenge ($S=2.41$), as well as their willingness to continue using the software for future learning ($S=2.54$). These results support the effectiveness of integrating game characteristics into educational software to overcome students' boredom with learning and improve their learning interest and efficiency. The findings highlight the potential of gamified learning approaches in vocational colleges to enhance students' learning experiences and promote their team organization and communication skills. Furthermore, the close correspondence between the numerical value of system reliability derived from the game and the results obtained from mathematical derivation underscores the educational value of the learning software. The positive feedback from students further validates the software's effectiveness in enhancing their interest and enthusiasm for learning abstract concepts and complex formulas.

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